nature portfolio

Corresponding author(s): Rafael Blanco-Dominguez
Bruno Silva-Santos

Last updated by author(s): Oct 2, 2024

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our Editorial Policies and the Editorial Policy Checklist.

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed	
☐	☒	The exact sample size (n) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒	A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒	The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☒	☐	A description of all covariates tested
☐	☒	A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒	A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒	For null hypothesis testing, the test statistic (e.g. F, t, r) with confidence intervals, effect sizes, degrees of freedom and P value noted *Give P values as exact values whenever suitable.*
☒	☐	For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐	For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐	Estimates of effect sizes (e.g. Cohen's d, Pearson's r), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	No custom code was used to collect the data BD FACSDIVA software version 9.0 was used for collection of flow cytometry data.
Data analysis	All analyses are described in the relevant section of Methods. For flow cytometry data, data were analyzed with Flowjo software v10.10.0. The FlowJo plugin X-shift (version 1.4.1) was used to classify endogenous Vδ1 T cells in different clusters. For statistical analyses and graphs, GraphPad Prism v8.4.0

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

This study includes no data deposited in external repositories. Source data are provided with this paper. All cancer patient and TCGA data were obtained from the publicly available official websites of the project (http://cancergenome.nih.gov and https://portal.gdc.cancer.gov/). Additional data supporting the findings of this study are available from the corresponding authors on reasonable request.

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	Sex and gender are not considered in the study design of human samples, which includes biological male and female sexes.
Reporting on race, ethnicity, or other socially relevant groupings	We did not consider race, ethnicity or other socialy relevant grouping in our analyses.
Population characteristics	Both colon and rectal cancer patients ranged between 52 and 85 years old and were not subjected to any treatment.
Recruitment	Colon and rectal tumor specimens from patients were prospectively collected by colonoscopy (at the time of tumor ressection)and prior to treatment between January 2023 and March 2024 from the Hospital Santa María (Lisbon, Portugal). In addition, a blood sample was collected from the same patients and PBMCs were cryopreserved at the biobank of the Instituto de Medicina Molecular João Lobo Antunes. In parallel, blood samples from 11 randomized age-matched healthy donors from the biobank were analyzed. Seven patient-derived organoids and tumor-infiltrating leukocytes expanded from CRC patients (four MMRp/MSS and three MMRd/MSI) at the Leiden University Medical Center as described below were included in this work. Patient samples were anonymized and handled according to the medical ethical guidelines described in the Code of Conduct for Proper Secondary Use of Human Tissue of the Dutch Federation of Biomedical Scientific Societies.
Ethics oversight	This investigator-initiated study was conducted in accordance with the principles of the Declaration of Helsinki and the International Conference on Harmonization Good Clinical Practice guidelines and was approved by the ethics committee of Centro Académico de Medicina de Lisboa (no. 329/20 and 329/20/21A). All patients included in the study provided written informed consent before sample collection. Participants were not compensated for study participation.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	Sample size was determined throughout the manuscript based on our experimental observation and experience to guarantee reliable and reproducible results. In the case of human samples, size of experiments and analyses was determined by the availability of recruited patient samples at the time of analysis. Sample size is indicated in the figure legends accordingly.
Data exclusions	No data were excluded from analyses
Replication	At least 3 biological and/or technical replicates were used for all our experiment as it is described in the figure legends. For in vivo experiments and patient samples, biological replicates are always represented. For in vitro killing assays, biologically replicated DOT donors are used For induction of ligand expression on tumor cell lines, technical replicates were used.
Randomization	Mice were randomly assigned to the different groups.
Blinding	Data collection and analysis were not performed blind to the conditions of the experiments

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☐	☒ Antibodies
☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☐	☒ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☐	☒ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used

Marker /Fluorochrome / Clone / Company / Reference /Dilution Factor
B7-H6 PE 875001 R&D Systems FAB144P-100 15
CD107a BV605 H4A3 Biolegend 328634 200
CD158 (KIR2D) BUV395 HP-MA4 BD 567326 200
CD3 BUV805 SK7 BD 612894 200
CD3 BV510 OKT3 Biolegend 317332 200
CD45 PE-CF594 HI30 BD 562279 400
CD45 AF700 HI30 Biolegend 304023/4 200
CD45 PEDzzle 2D1 Biolegend 368529/30 200
CD69 RB744 FN50 BD 570501 100
CD69 BV605 FN50 Biolegend 310937/8 100
CD69 PERCPCy5.5 FN50 BioLegend 310925 100
CD94 PE HP-3D9 BD 555889 100
CD96 APC NK92.39 BioLegend 338409 100
CD96 BV421 6F9 BD 742794 100
CD96 PercpCy5.5 NK92.39 BioLegend 338411 100
CTLA4//CD152 RY586 BNI3 BD 568442 100
DNAM1 FITC 11A8 BioLegend 338304 100
DNAM-1 BV711 11A8 BioLegend 338333 100
Gzmb Pacific blue GB11 BioLegend 515408 200
IFNg APC 4S.B3 Invitrogen 51-7319-42 100
IFNg APC 4S.B3 BioLegend 502512 100
IFNg PercpCy5.5 4S.B3 BioLegend 502526 100
KLRG1 RB780 Z7-205.rMAb BD 569137 100
LAG-3 APC-R700 T47-530 BD 565775 100
LAG-3 APC 3DS223H Invitrogen 17-2239-42 100
MICA/B BV711 6D4 BD 742326 4
Nectin-2 APC TX31 BioLegend 337412 20
Nectin-2 Percp-Cy5.5 TX31 BioLegend 337415 20
NKG2a BB700 131411 BD 747926 100
NKG2D BV421 1D11 BioLegend 320822 100
NKG2D BV785 1D11 BioLegend 320830 100
NKG2D PECy7 1D11 BioLegend 320812 100
NKp30 BUV737 P30-15 BD 749128 100
Nkp30 APC P30-15 BioLegend 325210 100
NKp44 BUV615 P44-8 BD 752353 100
NKp46 AF647 9.00E+02 Biolegend 331910 100
PD1 APCCy7 EH12.2H7 BioLegend 329921 100
PD1 BV480 EH12.1 BD 566112 50
PD1 FITC MIH4 Invitrogen 11-9969-42 50
PD-L1 APC 29E.2A3 BD 568315 100
Perforin BV711 dG9 Sony 2140645 100
PVR PECy7 SKII.4 BioLegend 337614 40
TIGIT PECy7 A15153G BioLegend 372713 200
TIGIT PercpCy5.5 A15153G BioLegend 372717 100
TIM3 BV650 7D3 BD 565564 100
TNF BUV395 Mab11 BD 563996 200
TNF PECy7 Mab11 Biolegend 502930 200
ULBP-1 PERCP 170818 R&D Systems FAB1380C 5
ULBP-2/5/6 BV605 165903 BD 748131 7

ULBP3 AF405 166510 R&D Systems FAB1517V 5
ULBP-4 AF750 709116 R&D Systems FAB6285S 15
Vd1 PE REA173 Miltenyi 130-120-440 200
Vd1 FITC REA173 Miltenyi 130-118-362 100
Vd1 APC REA173 Miltenyi 130-119-145 200
Vd2 APC-Vio770 REA771 Miltenyi 130-111-130 200
IgG1 BV785 MOPC-21 Biolegend 400169 equimolar
IgG1 BV711 MOPC-21 Biolegend 400167 equimolar
IgG1 BV421 X40 BD 400157 equimolar
IgG1 BV650 X40 BD 563231 equimolar
IgG1 FITC MOPC-21 BioLegend 400108 equimolar
IgG1 RB744 X40 BD 570519 equimolar
IgG1 BV605 MOPC-21 BioLegend 400161 equimolar
IgG1 APC-R700 X40 BD 564974 equimolar
IgG1 BUV737 X40 BD 612758 equimolar
IgG1 PE MOPC-21 BioLegend 400112 equimolar
IgG1 BUV615 X40 BD 612986 equimolar
IgG1 RB780 X40 BD 568532 equimolar
IgG1 PercpCy5.5 MOPC-21 Biolegend 400149 equimolar
IgG1 PECy7 MOPC-21 Biolegend 400125 equimolar
IgG2a PECy7 MOPC-173 BioLegend 400232 equimolar
IgG2a BB700 G155-178 BD 566419 equimolar
IgG2a RY586 G155-178 BD 568131 equimolar
IgG2a AF405 20102 R&D Systems IC003V equimolar
IgG2a BV605 G155-178 BD 562778 equimolar
IgG2a BV711 G155-178 BD 563345 equimolar
IgG2b BUV395 27-35 BD 563558 equimolar
IgG2b APC MPC-11 BioLegend 400322 equimolar
IgG2b AF750 133303 R&D Systems IC0041S equimolar

Mouse CD11b APCCy7 M1/70 BioLegend 101226 200
Mouse CD11b BV711 M1/70 BioLegend 101241 200
mouseCD45 PEDdzzle 30-F11 BioLegend 103146 200
mouseCD45 BV510 30-F11 BioLegend 103137 200

Zombie Violet BioLegend 423114 500
Zombie Acqua BioLegend 423102 500
Live/Dead NIR Invitrogen L34976 5000
Annexin V AF647 BioLegend 640912 100
Caspase 3/7 Green Invitrogen R37111 50
CFSE Invitrogen C34554 20000
CellTraceViolet Invitrogen C34557 10000
CellTraceYellow Invitrogen C34567 10000

Isotype controls were used at equimolar concentrations of the marker antibody

Validation

Antibodies were used following the manufacturer's instructions.

The antibodies used in this study were used according to the manufacturer's recommendation. Validation was based on the description provided on the manufacturer's homepage.
Marker /Fluorochrome/ Clone /Company /Reference/Link
B7-H6 PE B7-H6 PE 875001 FAB7144P-100 R&D Systems https://www.rndsystems.com/products/human-b7-h6-pe-conjugated-antibody-875001_fab7144p
CD107a BV605 H4A3 Biolegend 328634 https://www.biolegend.com/de-de/products/brilliant-violet-605-anti-human-cd107a-lamp-1-antibody-8975
CD158 (KIR2D) BUV395 HP-MA4 BD 567326 https://www.bdbiosciences.com/en-eu/products/reagents/flow-cytometry-reagents/research-reagents/single-color-antibodies-ruo/buv395-mouse-anti-human-kir2dl1-s1-s3-s5-cd158.567326?tab=product_details
CD3 BUV805 SK7 BD 612894 https://www.bdbiosciences.com/en-eu/products/reagents/flow-cytometry-reagents/research-reagents/single-color-antibodies-ruo/buv805-mouse-anti-human-cd3.612894?tab=product_details
CD3 BV510 OKT3 Biolegend 317332 https://www.biolegend.com/de-de/products/brilliant-violet-510-anti-human-cd3-antibody-8009
CD45 PE-CF594 HI30 BD 562279 https://www.bdbiosciences.com/en-eu/products/reagents/flow-cytometry-reagents/research-reagents/single-color-antibodies-ruo/pe-cf594-mouse-anti-human-cd45.562279?tab=product_details
CD45 AF700 HI30 Biolegend 304023/4 https://www.biolegend.com/de-de/products/alexa-fluor-700-anti-human-cd45-antibody-3401
CD45 PEDzzle 2D1 Biolegend 368529/30 https://www.biolegend.com/de-de/products/pe-dazzle-594-anti-human-cd45-antibody-14705
CD69 RB744 FN50 BD 570501 https://www.bdbiosciences.com/en-eu/products/reagents/flow-cytometry-reagents/research-reagents/single-color-antibodies-ruo/rb744-mouse-anti-human-cd69.570501?tab=product_details
CD69 BV605 FN50 Biolegend 310937/8 https://www.biolegend.com/de-de/products/brilliant-violet-605-anti-human-cd69-antibody-8704
CD69 PERCPCy5.5 FN50 BioLegend 310925 https://www.biolegend.com/de-de/products/percp-cyanine5-5-anti-human-cd69-antibody-5606
CD94 PE HP-3D9 BD 555889 https://www.bdbiosciences.com/en-eu/products/reagents/flow-cytometry-reagents/research-reagents/

single-color-antibodies-ruo/pe-mouse-anti-human-cd94.555889?tab=product_details
CD96 APC NK92.39 BioLegend 338409 https://www.biolegend.com/de-de/products/apc-anti-human-cd96-tactile-antibody-12180
CD96 BV421 6F9 BD 742794 https://www.bdbiosciences.com/en-eu/products/reagents/flow-cytometry-reagents/research-reagents/single-color-antibodies-ruo/bv421-mouse-anti-human-cd96.742794?tab=product_details
CD96 PercpCy5.5 NK92.39 BioLegend 338411 https://www.biolegend.com/de-de/products/percp-cyanine5-5-anti-human-cd96-tactile-antibody-12181
CTLA4//CD152 RY586 BNI3 BD 568442 https://www.bdbiosciences.com/en-eu/products/reagents/flow-cytometry-reagents/research-reagents/single-color-antibodies-ruo/ry586-mouse-anti-human-cd152.568442?tab=product_details
DNAM1 FITC 11A8 BioLegend 338304 https://www.biolegend.com/de-de/products/fitc-anti-human-cd226-dnam-1-antibody-5544
DNAM-1 BV711 11A8 BioLegend 338333 https://www.biolegend.com/de-de/products/brilliant-violet-711-anti-human-cd226-antibody-15592
Gzmb Pacific blue GB11 BioLegend 515408 https://www.biolegend.com/de-de/products/pacific-blue-anti-human-mouse-granzyme-b-antibody-8612
IFNg APC 4S.B3 Invitrogen 17-7319-82 https://www.thermofisher.com/antibody/product/IFN-gamma-Antibody-clone-4S-B3-Monoclonal/17-7319-82
IFNg APC 4S.B3 BioLegend 502512 https://www.biolegend.com/de-de/products/apc-anti-human-ifn-gamma-antibody-1012
IFNg PercpCy5.5 4S.B3 BioLegend 502526 https://www.biolegend.com/de-de/products/percp-cyanine5-5-anti-human-ifn-gamma-antibody-4426
KLRG1 RB780 Z7-205.rMAb BD 569137 https://www.bdbiosciences.com/en-eu/products/reagents/flow-cytometry-reagents/research-reagents/single-color-antibodies-ruo/rb780-mouse-anti-human-klrg1.569137?tab=product_details
LAG-3 APC-R700 T47-530 BD 565775 https://www.bdbiosciences.com/en-eu/products/reagents/flow-cytometry-reagents/research-reagents/single-color-antibodies-ruo/apc-r700-mouse-anti-human-lag-3-cd223.565775?tab=product_details
LAG-3 APC 3DS223H Invitrogen 17-2239-42 https://www.thermofisher.com/antibody/product/CD223-LAG-3-Antibody-clone-3DS223H-Monoclonal/17-2239-42
Mouse CD11b APCCy7 M1/70 BioLegend 101226 https://www.biolegend.com/de-de/products/apc-cyanine7-anti-mouse-human-cd11b-antibody-3930
Mouse CD11b BV711 M1/70 BioLegend 101241 https://www.biolegend.com/de-de/products/brilliant-violet-711-anti-mouse-human-cd11b-antibody-7927
MICA/B BV711 6D4 BD 742326 https://www.bdbiosciences.com/en-eu/products/reagents/flow-cytometry-reagents/research-reagents/single-color-antibodies-ruo/bv711-mouse-anti-human-mic-a-b.742326?tab=product_details
mouseCD45 PEDdzzle 30-F11 BioLegend 103146 https://www.biolegend.com/de-de/products/pe-dazzle-594-anti-mouse-cd45-antibody-10070
mouseCD45 BV510 30-F11 BioLegend 103137 https://www.biolegend.com/de-de/products/brilliant-violet-510-anti-mouse-cd45-antibody-7995
Nectin-2 APC TX31 BioLegend 337412 https://www.biolegend.com/de-de/products/apc-anti-human-cd112-nectin-2-antibody-11898
Nectin-2 Percp-Cy5.5 TX31 BioLegend 337415 https://www.biolegend.com/de-de/products/percp-cyanine5-5-anti-human-cd112-nectin-2-antibody-11881
NKG2a BB700 131411 BD 747926 https://www.bdbiosciences.com/en-eu/products/reagents/flow-cytometry-reagents/research-reagents/single-color-antibodies-ruo/bb700-mouse-anti-human-nkg2a-cd159a.747926?tab=product_details
NKG2D BV421 1D11 BioLegend 320822 https://www.biolegend.com/de-de/products/brilliant-violet-421-anti-human-cd314-nkg2d-antibody-12395
NKG2D BV785 1D11 BioLegend 320830 https://www.biolegend.com/de-de/products/brilliant-violet-785-anti-human-cd314-nkg2d-antibody-15537
NKG2D PECy7 1D11 BioLegend 320812 https://www.biolegend.com/de-de/products/pe-cyanine7-anti-human-cd314-nkg2d-antibody-6499
NKp30 BUV737 P30-15 BD 749128 https://www.bdbiosciences.com/en-eu/products/reagents/flow-cytometry-reagents/research-reagents/single-color-antibodies-ruo/buv737-mouse-anti-human-cd337-nkp30.749128?tab=product_details
Nkp30 APC P30-15 BioLegend 325210 https://www.biolegend.com/de-de/products/apc-anti-human-cd337-nkp30-antibody-3856
NKp44 BUV615 P44-8 BD 752353 https://www.bdbiosciences.com/en-eu/products/reagents/flow-cytometry-reagents/research-reagents/single-color-antibodies-ruo/buv615-mouse-anti-human-nkp44-cd336.752353?tab=product_details
NKp46 AF647 9E2 Biolegend 331910 https://www.biolegend.com/de-de/products/alexa-fluor-647-anti-human-cd335-nkp46-antibody-4578
PD1 APCCy7 EH12.2H7 BioLegend 329921 https://www.biolegend.com/de-de/products/apc-cyanine7-anti-human-cd279-pd-1-antibody-7121
PD1 BV480 EH12.1 BD 566112 https://www.bdbiosciences.com/en-eu/products/reagents/flow-cytometry-reagents/research-reagents/single-color-antibodies-ruo/bv480-mouse-anti-human-cd279-pd-1.566112?tab=product_details
PD1 FITC MIH4 Invitrogen 11-9969-42 https://www.thermofisher.com/antibody/product/CD279-PD-1-Antibody-clone-MIH4-Monoclonal/11-9969-42
PD-L1 APC 29E.2A3 BD 568315 https://www.bdbiosciences.com/en-eu/products/reagents/flow-cytometry-reagents/research-reagents/single-color-antibodies-ruo/apc-mouse-anti-human-pd-l1-cd274.568315?tab=product_details
Perforin BV711 dG9 Sony 2140645 https://www.sonybiotechnology.com/gb/brilliant-violet-711tm-anti-human-perforin
PVR PECy7 SKII.4 BioLegend 337614 https://www.biolegend.com/de-de/products/pe-cyanine7-anti-human-cd155-pvr-antibody-11925
TIGIT PECy7 A15153G BioLegend 372713 https://www.biolegend.com/de-de/products/pe-cyanine7-anti-human-tigit-vstm3-antibody-13951
TIGIT PercpCy5.5 A15153G BioLegend 372717 https://www.biolegend.com/de-de/products/percp-cyanine5-5-anti-human-tigit-vstm3-antibody-13948
TIM3 BV650 7D3 BD 565564 https://www.bdbiosciences.com/en-eu/products/reagents/flow-cytometry-reagents/research-reagents/single-color-antibodies-ruo/bv650-mouse-anti-human-tim-3-cd366.565564?tab=product_details
TNF BUV395 Mab11 BD 563996 https://www.bdbiosciences.com/en-eu/products/reagents/flow-cytometry-reagents/research-reagents/single-color-antibodies-ruo/buv395-mouse-anti-human-tnf.563996?tab=product_details
TNF PECy7 Mab11 Biolegend 502930 https://www.biolegend.com/de-de/products/pe-cyanine7-anti-human-tnf-alpha-antibody-6515
ULBP-1 PERCP 170818 R&D Systems FAB1380C https://www.rndsystems.com/products/human-ulbp-1-percp-conjugated-antibody-170818_fab1380c
ULBP-2/5/6 BV605 165903 BD 748131 https://www.bdbiosciences.com/en-eu/products/reagents/flow-cytometry-reagents/research-reagents/single-color-antibodies-ruo/bv605-mouse-anti-human-ulbp-2-5-6.748131?tab=product_details

ULBP3 AF405 166510 R&D Systems FAB1517V https://www.rndsystems.com/products/human-ulbp-3-alexa-fluor-405-conjugated-antibody-166510_fab1517v
ULBP-4 AF750 709116 R&D Systems FAB6285S https://www.rndsystems.com/products/human-ulbp-4-raet1e-alexa-fluor-750-conjugated-antibody-709116_fab6285s
Vd1 PE REA173 Miltenyi 130-120-440 https://www.miltenyibiotec.com/US-en/products/tcr-vd1-antibody-anti-human-reafinity-rea173.html#Conjugate=PE:size=100-tests-in-200-uL
Vd1 FITC REA173 Miltenyi 130-118-362 https://www.miltenyibiotec.com/US-en/products/tcr-vd1-antibody-anti-human-reafinity-rea173.html#Conjugate=PE:size=100-tests-in-200-uL
Vd1 APC REA173 Miltenyi 130-119-145 https://www.miltenyibiotec.com/UN-en/products/tcr-vd1-antibody-anti-human-reafinity-rea173.html#conjugate=apc:size=100-tests-in-200-ul
Vd2 APC-Vio770 REA771 Miltenyi 130-111-130 https://www.miltenyibiotec.com/UN-en/products/tcr-vd2-antibody-anti-human-reafinity-rea771.html#Conjugate=APC-Vio-770:size=30-tests-in-60-uL
IgG1 BV785 MOPC-21 Biolegend 400169 https://www.biolegend.com/de-de/products/brilliant-violet-785-mouse-igg1-kappa-isotype-ctrl-7955
IgG1 BV711 MOPC-21 Biolegend 400167 https://www.biolegend.com/de-de/products/brilliant-violet-711-mouse-igg1-kappa-isotype-ctrl-7930
IgG1 BV421 X40 Biolegend 400157 https://www.biolegend.com/en-gb/products/brilliant-violet-421-mouse-igg1-kappa-isotype-ctrl-7194?GroupID=BLG15288
IgG1 BV650 X40 BD 563231 https://www.bdbiosciences.com/en-pt/products/reagents/flow-cytometry-reagents/research-reagents/flow-cytometry-controls-and-lysates/bv650-mouse-igg1-k-isotype-control.563231?tab=product_details
IgG1 FITC MOPC-21 BioLegend 400108 https://www.biolegend.com/de-de/products/fitc-mouse-igg1-kappa-isotype-ctrl-1406
IgG1 RB744 X40 BD 570519 https://www.bdbiosciences.com/en-pt/products/reagents/flow-cytometry-reagents/research-reagents/single-color-antibodies-ruo/rb744-mouse-igg1-isotype-control.570519?tab=product_details
IgG1 BV605 MOPC-21 BioLegend 400161 https://www.biolegend.com/de-de/products/brilliant-violet-605-mouse-igg1-kappa-isotype-ctrl-7630
IgG1 APC-R700 X40 BD 564974 https://www.bdbiosciences.com/en-pt/products/reagents/flow-cytometry-reagents/research-reagents/flow-cytometry-controls-and-lysates/apc-r700-mouse-igg1-isotype-control.564974?tab=product_details
IgG1 BUV737 X40 BD 612758 https://www.bdbiosciences.com/en-pt/products/reagents/flow-cytometry-reagents/research-reagents/flow-cytometry-controls-and-lysates/buv737-mouse-igg1-isotype-control.612758?tab=product_details
IgG1 PE MOPC-21 BioLegend 400112 https://www.biolegend.com/de-de/products/pe-mouse-igg1-kappa-isotype-ctrl-1408
IgG1 BUV615 X40 BD 612986 https://www.bdbiosciences.com/en-pt/products/reagents/flow-cytometry-reagents/research-reagents/flow-cytometry-controls-and-lysates/buv615-mouse-igg1-isotype-control.612986?tab=product_details
IgG1 RB780 X40 BD 568532 https://www.bdbiosciences.com/en-pt/search-results?searchKey=568532
IgG1 PercpCy5.5 MOPC-21 Biolegend 400149 https://www.biolegend.com/de-de/products/percp-cyanine5-5-mouse-igg1-kappa-isotype-ctrl-4205
IgG1 PECy7 MOPC-21 Biolegend 400125 https://www.biolegend.com/de-de/products/pe-cyanine7-mouse-igg1-kappa-isotype-ctrl-1926
IgG2a PECy7 MOPC-173 BioLegend 400232 https://www.biolegend.com/de-de/products/pe-cyanine7-mouse-igg2a-kappa-isotype-ctrl-1924
IgG2a BB700 G155-178 BD 566419 https://www.bdbiosciences.com/en-pt/products/reagents/flow-cytometry-reagents/research-reagents/flow-cytometry-controls-and-lysates/bb700-mouse-igg2a-isotype-control.566419?tab=product_details
IgG2a RY586 G155-178 BD 568131 https://www.bdbiosciences.com/en-pt/products/reagents/flow-cytometry-reagents/research-reagents/single-color-antibodies-ruo/ry586-mouse-igg2a-isotype-control.568131?tab=product_details
IgG2a AF405 20102 R&D Systems IC003V https://www.rndsystems.com/products/mouse-igg2a-alexa-fluor-405-conjugated-isotype-control_ic003v
IgG2a BV605 G155-178 BD 562778 https://www.bdbiosciences.com/en-pt/products/reagents/flow-cytometry-reagents/research-reagents/flow-cytometry-controls-and-lysates/bv605-mouse-igg2a-k-isotype-control.562778?tab=product_details
IgG2a BV711 G155-178 BD 563345 https://www.bdbiosciences.com/en-pt/products/reagents/flow-cytometry-reagents/research-reagents/flow-cytometry-controls-and-lysates/bv711-mouse-igg2a-isotype-control.563345?tab=product_details
IgG2b BUV395 27-35 BD 563558 https://www.bdbiosciences.com/en-pt/products/reagents/flow-cytometry-reagents/research-reagents/flow-cytometry-controls-and-lysates/buv395-mouse-igg2b-isotype-control.563558?tab=product_details
IgG2b APC MPC-11 BioLegend 400322 https://www.biolegend.com/de-de/products/apc-mouse-igg2b-kappa-isotype-ctrl-1410
IgG2b AF750 133303 R&D Systems IC0041S https://www.rndsystems.com/products/mouse-igg2b-alexa-fluor-750-conjugated-isotype-control_ic0041s

Eukaryotic cell lines

Policy information about cell lines and Sex and Gender in Research

Cell line source(s)

All CRC cell lines - COLO-320DM (ATCC CCL-220, female, STR profiling), DLD1 (ATCC CCL-221, male), HCT116 (ATCC CCL-247, male), HT-29 (ATCC HTB-38, female), SW480 (ATCC CCL-228, male) and SW620 (ATCC CCL-227, male) - were obtained from the American Type Culture Collection (ATCC). All cell lines were authenticated by ATCC by STR profiling.

Authentication

All CRC cell lines - COLO-320DM (ATCC CCL-220, female, STR profiling), DLD1 (ATCC CCL-221, male), HCT116 (ATCC CCL-247, male), HT-29 (ATCC HTB-38, female), SW480 (ATCC CCL-228, male) and SW620 (ATCC CCL-227, male) - were obtained from the American Type Culture Collection (ATCC). All cell lines were authenticated by ATCC by STR profiling.

Mycoplasma contamination

All cells were tested negative for mycoplasma contamination

Commonly misidentified lines
(See ICLAC register)

No commonly misidentified cell lines were used in the study

Animals and other research organisms

Policy information about studies involving animals; ARRIVE guidelines recommended for reporting animal research, and Sex and Gender in Research

Laboratory animals	Eight-to-fourteen weeks-old female non-obese diabetic mice transgenic for human IL-15 were acquired from Taconic (NOD.Cg-Prkdcscid Il2rgtm1Sug Tg(CMV-IL2/IL15)1-1Jic/JicTac). Mice were housed in rooms with a light-dark cycle of 14h/10h, temperature of 22-24 °C and relative humidity of 45-65% in SPF (specific pathogen free) animal rooms of the Gulbenkian Institute for Molecular Medicine Rodent Facility (Lisbon, Portugal).
Wild animals	The study does not involve wild animals
Reporting on sex	All the experiments were performed in female mice due to initial stock availability at the supplier. Although previous experience and pilot studies suggest a negligible effect of sex differences in these experiments, we cannot rule out a certain level of influence when using male mice.
Field-collected samples	The study does not involve field-collected samples.
Ethics oversight	Mouse experiments performed in this study were evaluated and approved by our institutional ethical committee (iMM-Orbea) and the national competent authority (DGAV) under the license number 0421/000/000/2023. Given that tumors grow intra-cecum, humane endpoints are defined based on the assessment of body condition and physical appearance and not on tumor size. In this study humane endpoints were not reached, since mice were euthanized for analysis before symptoms of disease were detected.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Plants

Seed stocks	n/a
Novel plant genotypes	n/A
Authentication	n/a

Flow Cytometry

Plots

Confirm that:

☒ The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).

☒ The axis scales are clearly visible. Include numbers along axes only for bottom left plot of group (a 'group' is an analysis of identical markers).

☒ All plots are contour plots with outliers or pseudocolor plots.

☒ A numerical value for number of cells or percentage (with statistics) is provided.

Methodology

Sample preparation	Single cell suspensions were incubated in PBS with fluorescently labelled monoclonal antibodies against the different surface markers for 20 minutes at 4 ºC in the dark. For intracellular markers (cytolytic granules and cytokines), cells were fixed and permeabilized for intracellular staining with Foxp3 staining buffer set (Invitrogen) following the manufacturer's instructions. When indicated, cells were stimulated with 0,2�/mL PMA (Merck) + 1�g/mL Ionomycin (Merck) in the presence of 10�g/mL Brefeldin A (Merck) + 0.1% Monensin (Invitrogen) for 3 hours at 37 ºC prior surface staining for the evaluation of functional intracellular markers.
Instrument	Flow cytometry acquisition was performed on a BD LSRFortessa X-20 Cell Analyzer (BD Biosciences), BD FACSymphony A5 SE (BD Biosciences) or Cytek Aurora (Cytek).
Software	Sample acquisition was performed with BD FACSDIVA software version 9.0. Data was analyzed with FlowJo 10 software (TreeStar). The FlowJo plugin X-shift was used to classify endogenous Vδ1 T cells in different clusters.
Cell population abundance	cell sorting was not used in this study

Gating strategy

For in vitro DOT cell analysis: FSC-A/SSC-A for lymphocytes, FSC-H/FSC-W for singlets, viability marker (neg) vs SSC-A for alive cells, and CD3+Vd1+cells. Then, phenotypic and functional markers were evaluated

For in vivo DOT cell infiltration/phenotype analyses: FSC-A/SSC-A for lymphocytes, FSC-H/FSC-W for singlets, viability marker (neg) vs SSC-A for alive cells, mouse-CD45(or CD11b) versus human CD45 for human lymphocytes (DOTs) and CD3+Vd1+cells. Then, phenotypic and functional markers were evaluated

For in vitro killing assay: FSC-A/SSC-A to exclude cell debris, FSC-H/FSC-W for singlets, CTV+ for tumor cells. Then apoptosis markers were evaluated

Isotype controls were used to determine the negative and positive population of the different markers.

☒ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.